11	The planet Jupiter has a radius of 7.15×10^7 m and is approximately 318 times more massive than the Earth. Given that the Earth has a radius of 6370 km, find the acceleration due to gravity on the surface of Jupiter.

	A	2.41×10^{-5} N kg^{-1}	**B**	2.73×10^{-3} N kg^{-1}	**C**	24.8 N kg^{-1}	**D**	273 N kg^{-1}

Answer: C

$$g = \frac{GM}{r^2}$$

$$g \propto \frac{M}{r^2}$$

Let the quantities with subscript '*J*' represent that of Jupiter while those with subscript '*E*' represent that of Earth.

$$\frac{g_J}{g_E} = \frac{M_J}{M_E}\left(\frac{r_E^{\,2}}{r_J^{\,2}}\right)$$

$$\frac{g_J}{g_E} = (318)\left(\frac{6370 \times 10^3}{7.15 \times 10^7}\right)^2$$

Since g_E = 9.81 N kg^{-1}, $g_J = (318)\left(\frac{6370 \times 10^3}{7.15 \times 10^7}\right)^2 (9.81)$

= 24.8 N kg^{-1}

12 A balloon contains 2.0 moles of helium gas at a temperature of 27 ºC. The molar mass of